CREATIVE TECHNO COLLEGE, ANGUL

Project Title: Simple Drawing App

Objective:

To create a basic graphical drawing tool that allows users to draw shapes or freehand lines using the mouse. This project helps students learn about event-driven programming, mouse handling, and canvas manipulation in Tkinter.

Key Features:

- Canvas area for drawing.
- Freehand drawing with mouse movement.
- Buttons to choose different tools (pen, eraser, shapes like line, rectangle, oval).
- Color selection (foreground and background).
- Clear canvas/reset button.
- Optional: Save drawing as an image.

How to Build It (Development Guidelines):

1. **Set Up the GUI:**
   - Create a `Canvas` widget to act as the drawing area.
   - Use `Button`, `Label`, and `ColorChooser` widgets for tool options.
2. **Implement Mouse Interactions:**
   - Bind mouse events like:
     - `<ButtonPress-1>`: Start drawing
     - `<B1-Motion>`: Draw while dragging
     - `<ButtonRelease-1>`: Stop drawing
   - Use `create_line()` or `create_oval()` for freehand and shapes.
3. **Add Drawing Tools:**
   - Pen: Basic line drawing.
   - Eraser: Draw white lines or background color.
   - Shapes: On button click, let users draw lines, rectangles, or ovals.
4. **Color and Size Customization:**
   - Use `colorchooser.askcolor()` for color picking.
   - Allow brush size selection via `Scale` widget.
5. **Optional Features:**
   - Add a "Save" button to save the canvas as an image using `PIL.ImageGrab`.

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- Add undo/redo functionality for advanced versions.

Technologies & Concepts to Use:

- **Python 3**
- **Tkinter** (Canvas, Events, Buttons, ColorChooser)
- **PIL** (Pillow library) — optional, for saving drawings
- Event handling and object tracking

Learning Outcomes:

- Understand canvas drawing and coordinate systems.
- Handle mouse events to interact with the GUI.
- Create dynamic and interactive GUI components.
- Optional: Work with external libraries (Pillow).

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